Review: *The Day convolution of two polynomial comonoids is the product internal category — in every Grothendieck topos*

Rick

To: Scot MacBeth (Kodamai) cc: Robin

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Reviewed:	`2026-10-22-day-convolution-topos-uniform.pdf`, content `b553b27`, 8 pp.; email UID 440
This review:	corresponds to WIP `0237120` (`grandpa-rick/work-in-progress`, `reviews/`)
Sources read:	arXiv abstracts of all five references; full text of Shapiro–Spivak 2305.00167 (SS), Gambino–Kock 0906.4931 (GK), Niu–Spivak 2312.00990 (NS). Lean file not run.

Counts: **3 MUST, 4 SHOULD, 6 NIT**.

Verdict

- **Theorem 1.1 is true, and §3 is sound internally.** No pointwise or Set-based assumption is used (your question (i); details in S2). It is also true in much greater generality than you state, and its Set case is already in print (M3, S1).
- **Theorem 1.2 is false, already for $\mathcal{E} = \mathbf{Set}$.** Comonoid morphisms in $\mathbf{Poly}_{\mathcal{E}}$ are *cofunctors*, not functors. In $\mathbf{Cat}^\sharp$ the product category is *not* the categorical product. Niu–Spivak say so explicitly. (M1)
- **The Gambino–Kock attribution is wrong.** (Your question (ii).) GK's distributive law is Π over Σ. Your ($\star$) is a finite-limit fact and needs no LCCC at all. The "LCCC + extensive" folklore you sharpen has no source. (M2)
- **§7 draws the line in the wrong place.** (Your question (iii).) Theorem 1.1 holds in any finitely complete $\mathcal{E}$ with exponentiable domain maps. That covers every elementary topos and every LCCC, with no coproducts needed. The one genuinely base-sensitive claim is the word "Day convolution" in the title. That is cited from a Set-only proposition and is unproved over $\mathcal{E}$. (M3)
- **Recommendation: not ready for publishable-result.** After the fixes, what remains is a correct, honestly credited short note (proposition level): "the Dirichlet tensor of internal categories is the product category, internally to any finite-limit category". I would read a second round gladly.

Findings

MUST

M1. Theorem 1.2, §4, abstract, §2.2 and Table 1: the universal property is false. SS Cor. 5.12 (checked in the full text) is an isomorphism between $\mathbf{Comon}(\mathbf{Poly}_{\mathcal{E}})$ and *internal cat-*

egories with exponentiable source and internal cofunctors (SS Def. 5.11). It is not an equivalence with $\mathbf{Cat}(\mathcal{E})$ (functors). So "the internal equivalence $\mathrm{cat}(\mathcal{E}) \simeq \mathbf{Cat}(\mathcal{E})$", and "since p_C is a functor, its forward image is a comonoid morphism" (§4), are both wrong. In $\mathrm{cat}(\mathcal{E})$ the product must be computed among cofunctors, and there it fails.

Literature. NS §8.2.4, opening sentences: "The usual product of categories is not the categorical product in $\mathbf{Cat}^\sharp$. It is, however, a monoidal product on $\mathbf{Cat}^\sharp$, coinciding with the parallel product $\otimes$." NS's chapter summary adds that the $\mathbf{Cat}^\sharp$-product of the walking arrow with itself has infinitely many objects.

Explicit witness ($\mathcal{E} = \mathbf{Set}$). Let $C = D = \mathbf{2} = \{0 \xrightarrow{a} 1\}$, and let E have objects e, x, z and non-identity arrows $u\colon e \to x$, $v\colon e \to z$. Define cofunctors $F, G\colon E \to \mathbf{2}$ by $e \mapsto 0$, $x, z \mapsto 1$, lifting a at e to u (for F) and to v (for G). The cofunctor laws hold trivially: there are no composable non-identity pairs. Any $H\colon E \to \mathbf{2} \times \mathbf{2}$ with $\pi_C H = F$ and $\pi_D H = G$ needs $H(e) = (0,0)$ and $H(x) = H(z) = (1,1)$. It must then lift $(a, \mathrm{id})\colon (0,0) \to (1,0)$ to an arrow out of e whose codomain maps to $(1,0)$. No such arrow exists. So $(c \otimes d, \pi_C, \pi_D)$ is not a product. (Your π_C, π_D *are* cofunctors, lifting $f \mapsto (f, \mathrm{id})$. The cone exists; it is the universality that fails.)

Fix. Delete Theorem 1.2, its row in Table 1, and the abstract sentence. Replace them with a remark: $\otimes$ is a monoidal, non-cartesian product on $\mathbf{Cat}^\sharp(\mathcal{E})$, generalizing NS Prop. 8.79. Also correct the description of SS in §2.2, §4 and Rem. 4.1: "isomorphism with internal categories *and cofunctors*". If you want a universal property, the honest one is in $\mathbf{Cat}(\mathcal{E})$ (functors), where $C \times D$ is the product by §4's first paragraph. But that is a statement about $\mathbf{Cat}(\mathcal{E})$, not about $\otimes$ on comonoids.

M2. §1 bullets, §2.1, §3 Step 2, §5 and Rem. 4.1: ($\star$) is not the Gambino–Kock law, and the "sharpening" is of an unsourced claim. GK §1.2, eq. (5)–(6), is the distributivity of *dependent products over dependent sums*: $\Pi_f \Sigma_u \cong \Sigma_v \Pi_g \Delta_e$. In type theory this is the "type-theoretic axiom of choice", $\prod \sum \cong \sum \prod$. It is what composition of polynomials uses. Your ($\star$), $\Sigma_{(i,k)} A_i \times B_k \cong \Sigma_i A_i \times \Sigma_k B_k$, says something else: the total space of a product of bundles is the product of total spaces. Concretely, at the point of use both sides are $(C_1 \times_{C_0} C_1) \times (D_1 \times_{D_0} D_1) \cong (C_1 \times D_1) \times_{C_0 \times D_0} (C_1 \times D_1)$. That holds in *any finitely complete* category: limits commute with limits. It needs no Π, no LCCC, and no colimit preservation. The gloss "$A \times (-) = \Delta_A^*$ is a left adjoint and so preserves the relevant colimits" is beside the point, because no colimit occurs. (Also, Δ_A^* lands in $\mathcal{E}/A$. The endofunctor $A \times (-)$ is $\Sigma_A \Delta_A$.)

On "LCCC + extensive": the only extensivity in GK is §4.3, for free monads and W-types (the coproduct $X + P(TX)$). It has nothing to do with $\triangleleft$ or $\otimes$. Unless you can quote a source that asserts extensivity is needed for $c \otimes d = C \times D$, there is nothing to sharpen. *Fix:* cite GK only for $\mathbf{Poly}(\mathcal{E})$ and $\triangleleft$ over an LCCC (that attribution is correct). Call ($\star$) "products commute with pullbacks". Delete the sharpening narrative from the abstract, §1 and §5.

M3. Title, abstract, Def. 2.2 and §7: wrong scope line; "Day convolution" is unproved over $\mathcal{E}$.

- *The theorem is far more general than stated.* SS work throughout in an arbitrary *finitely complete* $\mathcal{E}$: polynomials are exponentiable maps, $\triangleleft$ is defined (SS Thm. 3.26), $\otimes$ is the product of maps (SS Def. 4.1, preserved exponentiability by Lemma 4.3), normal duoidality holds (SS Thm. 4.5), and comonoids are internal categories (SS Thm. 5.6). Your §3 uses nothing beyond this. So Theorem 1.1 holds in every finitely complete $\mathcal{E}$ for internal categories with exponentiable dom. In particular it holds in every LCCC and every elementary topos, Grothendieck or not, with or without a natural numbers object. §7's "a bare LCCC without the requisite coproducts is not covered" is false. Dependent sums Σ_f are postcomposition and exist in any

category, and GK polynomials need no coproducts. The "open upper bound" is not open: it is the SS setting.

- *The base-sensitive part is the Day-convolution identification.* NS Prop. 3.79 is about $\mathbf{Set}^{\mathbf{Set}}$. Its proof is a co-Yoneda computation with coends in **Set**. Def. 2.2's "by Niu–Spivak this $\otimes$ is the Day convolution on $\mathbf{Poly}(\mathcal{E})$" is unsupported. Over $\mathcal{E}$ you would need to say Day convolution *of what*: $\mathcal{E}$-valued functors on $\mathcal{E}$? $\mathcal{E}$-indexed or enriched? And you would need the coends $\int^{A,B}$ over the large category $\mathcal{E}$, with copowers by hom-*sets* or internal homs, plus a pullback-stability argument. An elementary topos has no infinite coproducts, so this is exactly where Grothendieck vs. elementary could matter. Your proof never uses the Day description: $\otimes$ is *defined* as the product of bundles.
- *Fix.* Either (a) retitle to "the Dirichlet tensor", state Theorem 1.1 over a finitely complete $\mathcal{E}$ (exponentiable dom), and keep "Day convolution" only as the Set-level motivation via NS Prop. 3.79; or (b) prove the Day-convolution identification over Grothendieck $\mathcal{E}$ in a stated sense. Then that, and only that, is where the Grothendieck hypothesis lives. I recommend (a). Also rewrite §7 and Table 1 accordingly.

SHOULD

S1. Novelty and locators (Theorem 1.1, §1, §2.3). The Set case is NS **Prop. 8.79**: carrier $c \otimes d$, counit $\epsilon_c \otimes \epsilon_d$, comultiplication via the duoidal interchanger, and "codomains and composition are defined coordinate-wise". That is your Theorem 1.1 with $\mathcal{E} = \mathbf{Set}$, word for word. Cite it by number instead of "[3, the Set case]". Internally, SS do not state $c \otimes d = C \times D$, so the internal statement is fine to claim. But it is a short corollary of SS Thm. 4.5 and Thm. 5.6, and should be called one. SS's own technique gives a three-line alternative proof worth a remark: their Lemma 5.5 transports both comonoid and category structures along the fully faithful singular functor $\mathcal{E} \to \widehat{\mathcal{E}}$. This reduces to presheaves, and in presheaves the category side is pointwise. Locators: the interchanger is SS Thm. 4.5, eq. (16), with the Set formula in Ex. 4.6. Your "[4, Ex. 2.9, eq. (4)]" points at the example "$\mathrm{id}_{\mathcal{E}}$ is dense".

S2. §3 Step 3: internal validity (your question (i)). I find no smuggled assumption. No choice is used: identities and composition are given maps, and the only "choices" are chosen pullbacks. No excluded middle is used: there are no case splits and no decidability. No coend is computed pointwise: none is computed at all, since $\otimes$ is the product of maps. The element notation is legitimate as generalized elements, i.e. the internal language of a finite-limit category with exponentiable maps. But three points should be stated rather than glossed:

- The forward map of δ_c is not "cod". It is the Π_{dom}-transpose of cod, a map $C_0 \to \Sigma_{C_0}\Pi_{\mathrm{dom}}(C_0 \times C_1)$.
- On positions, ι is the canonical map $\Pi_X A \times \Pi_Y B \to \Pi_{X\times Y}(A \times B)$ (fibrewise $A^X \times B^Y \to (A \times B)^{X\times Y}$). Say in one line that it sends the pair of transposes of $\mathrm{cod}_C, \mathrm{cod}_D$ to the transpose of $\mathrm{cod}_C \times \mathrm{cod}_D$ (naturality of transposition).
- "On the nose" means equality in $\mathbf{Poly}_{\mathcal{E}}$, whose morphisms are *isomorphism classes* of diagrams (SS Def. 3.7, 3.13). That is what makes "=" meaningful despite the canonical isomorphism in the composition bullet. Rephrase "there is no coherence isomorphism to chase". Likewise, Rem. 3.1's "laws inherited from $C \times D$" needs the fact that SS's proof of Thm. 5.6 matches the *equations* one by one on unconstrained data. Their statement alone is a bijection of lawful structures. Cite the proof.

S3. §1, Lucyshyn-Wright [2]. The abstract of 2502.18557 is about functor categories and bifunctors over *non-symmetric* bases V, via graded categories. Nothing in it says "Day-convolution closure is special to Set-like bases". Quote a locator that does, or delete the sentence. The same goes for the Vec paragraph: "Vec is not LCCC" is true, but SS need only finite limits and exponentiable maps, so "relocates the Vec obstruction precisely" overstates.

S4. §6, Graph witness. This is a fine finite sanity check of Theorem 1.1 in a presheaf topos. But in a presheaf topos the category side is computed pointwise, so it cannot test the "non-pointwise" worry. The global-element point needs no test: id is a morphism, and global elements never enter. Present it as an illustration. Drop "to the author's knowledge, the first Lean certificate...". Give the Lean file's URL (the PDF has only a path).

NIT

N1. Def. 2.1: $p(X) = \Sigma_{A_p}\Pi_{s_p}A_p^*X$ is ill-typed. Π_{s_p} needs an object over B_p, so the formula should be $\Sigma_{A_p}\Pi_{s_p}B_p^*X$ (i.e. $\Delta_{B_p}X = X \times B_p$).

N2. §6: C_{pt} is a baseline (positive) control, not a "negative control". The negative controls are `bogus` and `notcat`.

5, Prop. 5.2 for the comonoid phrasing: I did not verify this locator. SS Thm. 5.6 with Rem. 5.7 is a safer citation.

N3. §1 says "Theorem 1.1 — ours" while also crediting "[3, the Set case]". Make it consistent once S1 is done.

N4. Table 1, row 1: the column "holds in" should say "any finitely complete $\mathcal{E}$ (exponentiable maps)". See M3.

N5. SS eq. (16) itself has a typo ($p_2 \triangleleft q_2$ for $p_2 \otimes q_2$). If you reproduce it, don't copy that.

What I checked, plainly

- **SS (2305.00167), full text.** Abstract: finite-limit $\mathcal{E}$, $\otimes$, duoidality, comonoids = internal categories with exponentiable source. Thm. 5.6 is the bijection of structures. Cor. 5.12 is the isomorphism with *cofunctors*. Your locators 5.6 and 5.12 are the right numbers, but the content is misdescribed (M1). Ex. 2.9 is the wrong locator (S1).
- **GK (0906.4931).** Polynomial functors over LCCCs, the framed bicategory, free monads. The distributive law is eq. (5)–(6), Π over Σ. Extensivity appears only in §4.3 (M2).
- **NS (2312.00990).** Prop. 3.79 is Day convolution on $\mathbf{Set}^{\mathbf{Set}}$ only. §8.2.4 has Prop. 8.79 = the Set case of Theorem 1.1, and the explicit statement that $\times$ is not the $\mathbf{Cat}^\sharp$-product.
- **Lucyshyn-Wright (2502.18557)** and **SGF (2111.10968)**: abstracts only. Titles and authors match your bibliography.